Genre in Grammar:
PCA on Sanskrit Morphological Variation

Linguistic Data 3: Data Modeling in ILs

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Introduction

- Sanskrit literary tradition spans over 2,400 years and encompasses five major genre categories—Vedic, Epic, Kavya, Shastra, and Puranic—whose morphological distinctiveness has never been quantitatively established at scale.

- In this project, we perform large-scale computational analysis using the Digital Corpus of Sanskrit (DCS), comprising 5.69 million tokens across 270 texts.

Research Questions

1. Do the five Sanskrit literary genres occupy distinct regions of a combined morphological–syntactic feature space, and can PCA reveal the principal axes of variation?

2. Which morphological and syntactic features contribute most to cross-genre separation, as indicated by PCA loadings?

3. Where unsupervised clusters diverge from the traditional genre taxonomy, what does qualitative evidence reveal about genre boundaries?

4. How do SVM, Logistic Regression, and LDA compare in genre classification accuracy when trained on PCA-reduced representations?

Initial Hypotheses

It is hypothesized that:

(i) PC1 will separate nominal-heavy registers (Shastra, Kavya) from verbal-heavy ones (Epic, Vedic) along a nominalization axis;

(ii) PC2 will capture compound density, separating Kavya (highly compound-heavy) from other genres;

(iii) Vedic texts will form a morphologically isolated cluster due to archaic case usage and verbal morphology distinct from Classical Sanskrit; and

(iv) SVM with a linear kernel will outperform LDA and Logistic Regression in genre classification, given the high-dimensional, linearly separable nature of morphological feature spaces.

Data

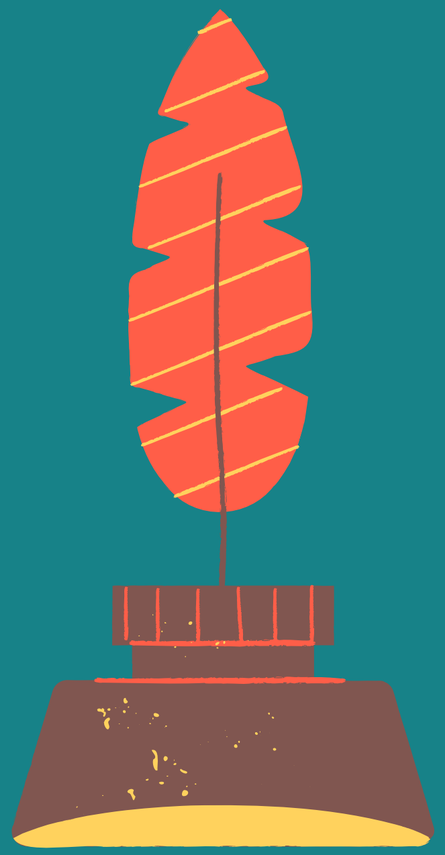
The DCS provides 5.6 million tokens across 270 text folders spanning 500 BCE to 1900 CE, with 10 columns per token like token ID, surface form, lemma, UPOS, XPOS, morphological features (FEATS), syntactic head (HEAD), dependency relation (DEPREL) etc.

Table 1: DCS corpus statistics by genre.

Genre	Texts	Chapters	Tokens	Treebank
Epic	4	2,735	1,581,085	42
Vedic	76	7,343	1,316,026	1,692
Kavya	47	367	672,774	3
Shastra	126	4,308	1,380,603	25
Puranic	17	1,147	737,928	4
Total	270	15,900	5,688,416	1,766

Genre labels were assigned via a curated mapping of all 270 DCS text folder names

Feature Extraction



Each text is represented as a 30-dimensional feature vector, all rates normalized per 1,000 tokens

Feature Category	Count	Specific Features / Description
POS Rates	8	NOUN, VERB, ADJ, ADP, PRON, NUM, ADV, PART
Case Rates	8	Nom, Acc, Inst, Dat, Abl, Gen, Loc, Voc (Nominal tokens only)
Tense Rates	4	Pres, Past, Fut, Imp (Verbal tokens only)
Mood Rates	4	Ind, Opt, Imp, Sub (Verbal tokens only)
Syntactic Features	3	Avg. dependency arc length, subordination rate, verb-final rate*
Nominal-to-Verbal	1	$(NOUN + ADJ + PRON)/\max(VERB, 1)$
Complexity (MCI)	1	Shannon entropy: $-\sum p_i \log_2 p_i$
Specific Markers	1	Compound rate (tokens with Case=Cpd)
Mantra Rate	1	IsMantra=True tokens per 1,000

*Available only for 1,766 Vedic Treebank chapters.

Methods

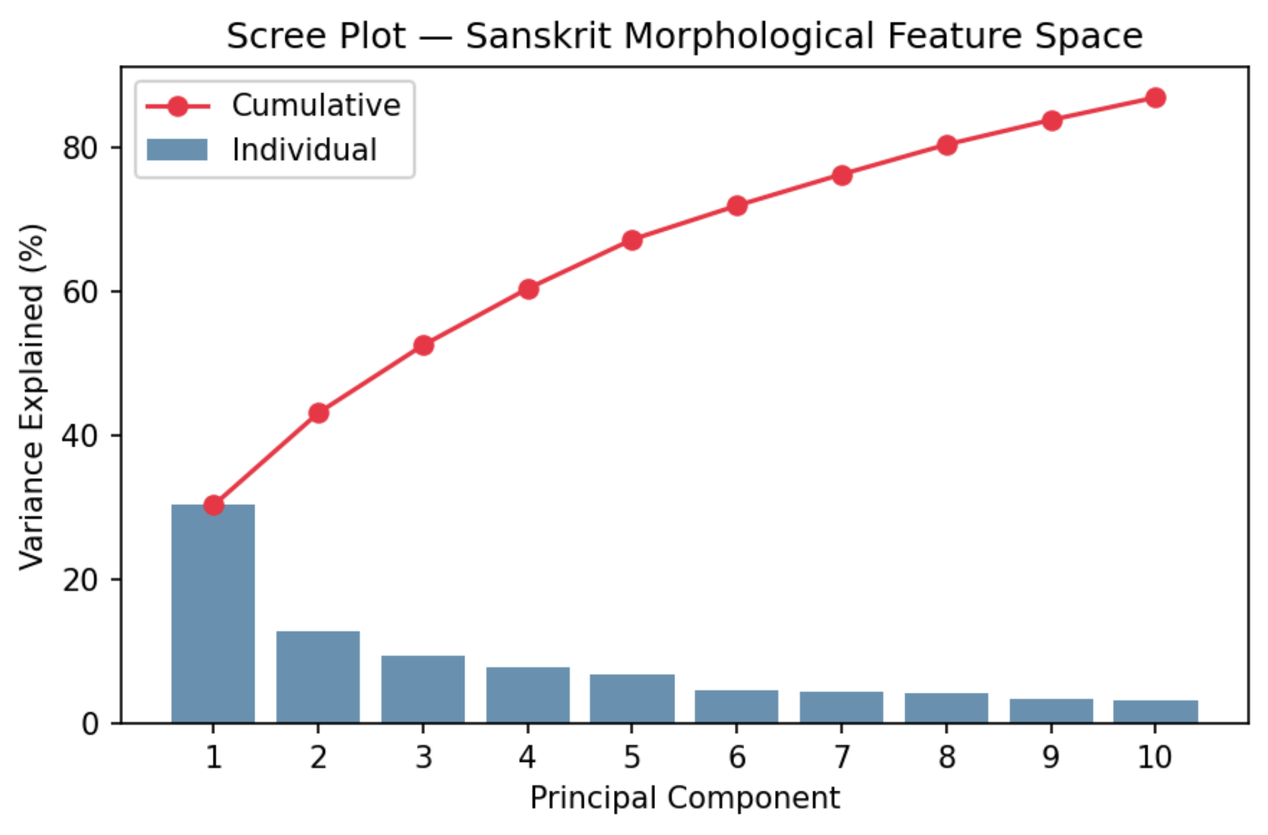
- **Dimensionality Reduction (PCA)**: Features were standardized to zero mean and unit variance before performing Principal Components Analysis (PCA). Six components (PC1–PC6) were retained, cumulatively explaining 72% of the variance.

- **Unsupervised Clustering**: K-means and Ward hierarchical clustering were applied to the PC scores and evaluated using the Adjusted Rand Index (ARI).

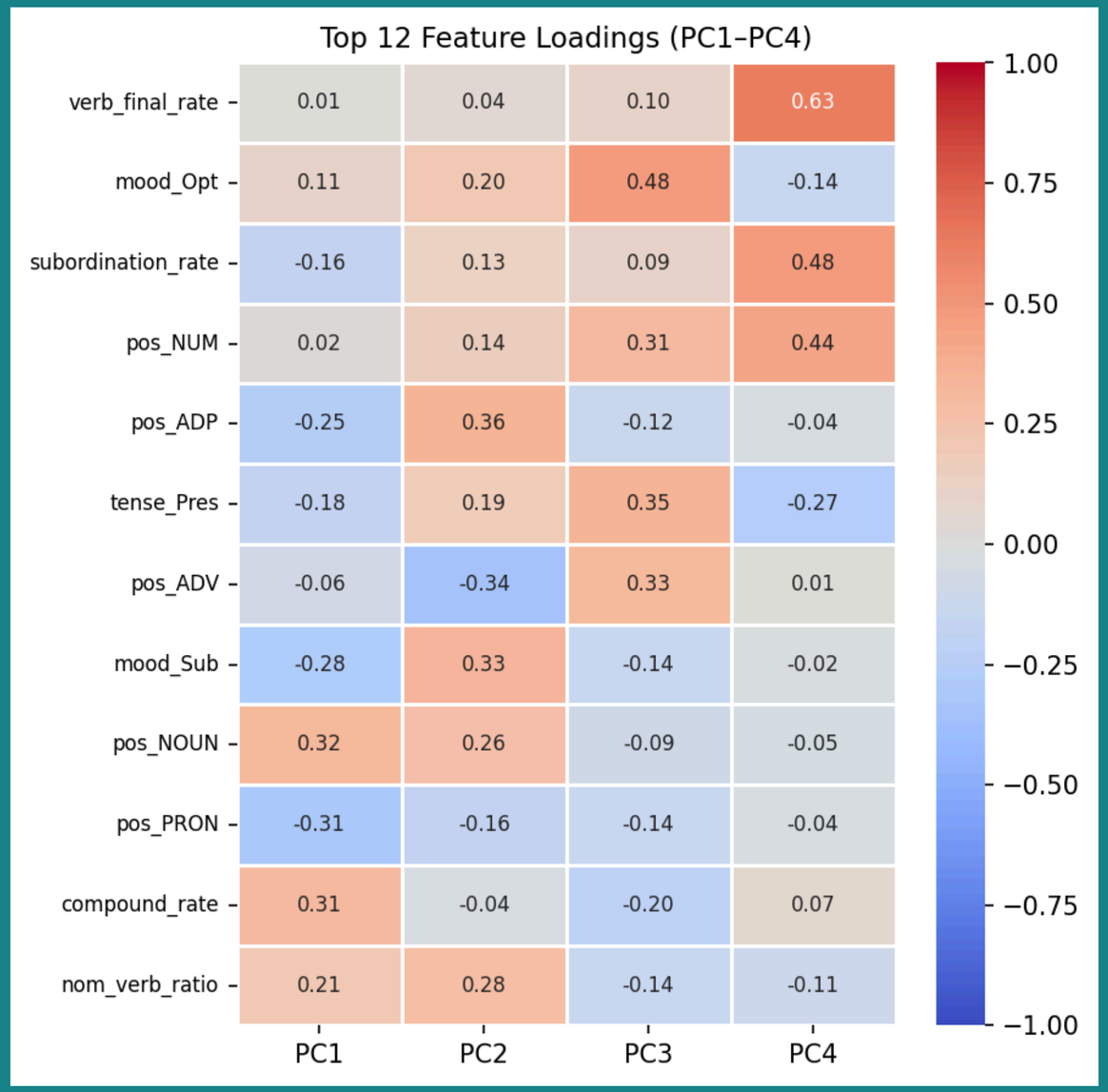
- **Supervised Classification**: SVM, Logistic Regression, and LDA were trained on the six components using 4-fold stratified cross-validation.

- **Diachronic Analysis**: Spearman rank correlations were calculated between text periods (BCE/CE) and morphological features to identify historical shifts.

PCA



PC1 explains 30.3% of variance, with subsequent PCs contributing 12.8%, 9.4%, 7.9%, 6.8%, and 4.7% respectively. The gradual elbow—rather than a sharp drop—indicates that Sanskrit genre variation is genuinely multidimensional: no single binary contrast accounts for the majority of morphological variance.



- **PC1: Nominalization Axis**

  It **confirms hypothesis 1** by separating verbal-heavy vedic texts (high in pronouns, verbs) from nominal-heavy Shastra and Kavya Texts.

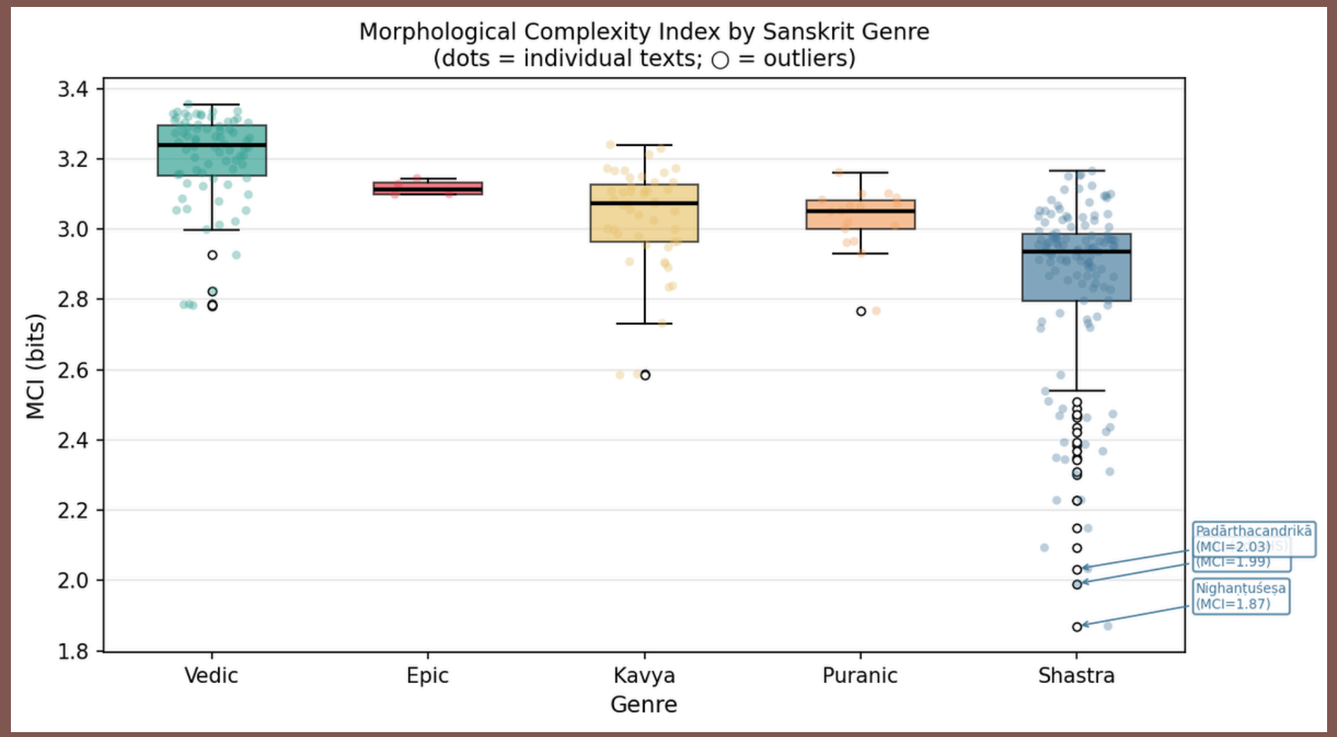
- **PC2: Syntactic Complexity Axis**

  It does not primarily capture compound density as predicted: compound density instead covaries with nominalization on PC1. **Hypothesis 2 is not confirmed.**

- **PC3: Archaic Verbal Mood Axis**

  Captures the Vedic–Classical divide at a finer grain, dominated by the optative mood, which is a defining marker of archaic Vedic Sanskrit.

Morphological Complexity Index



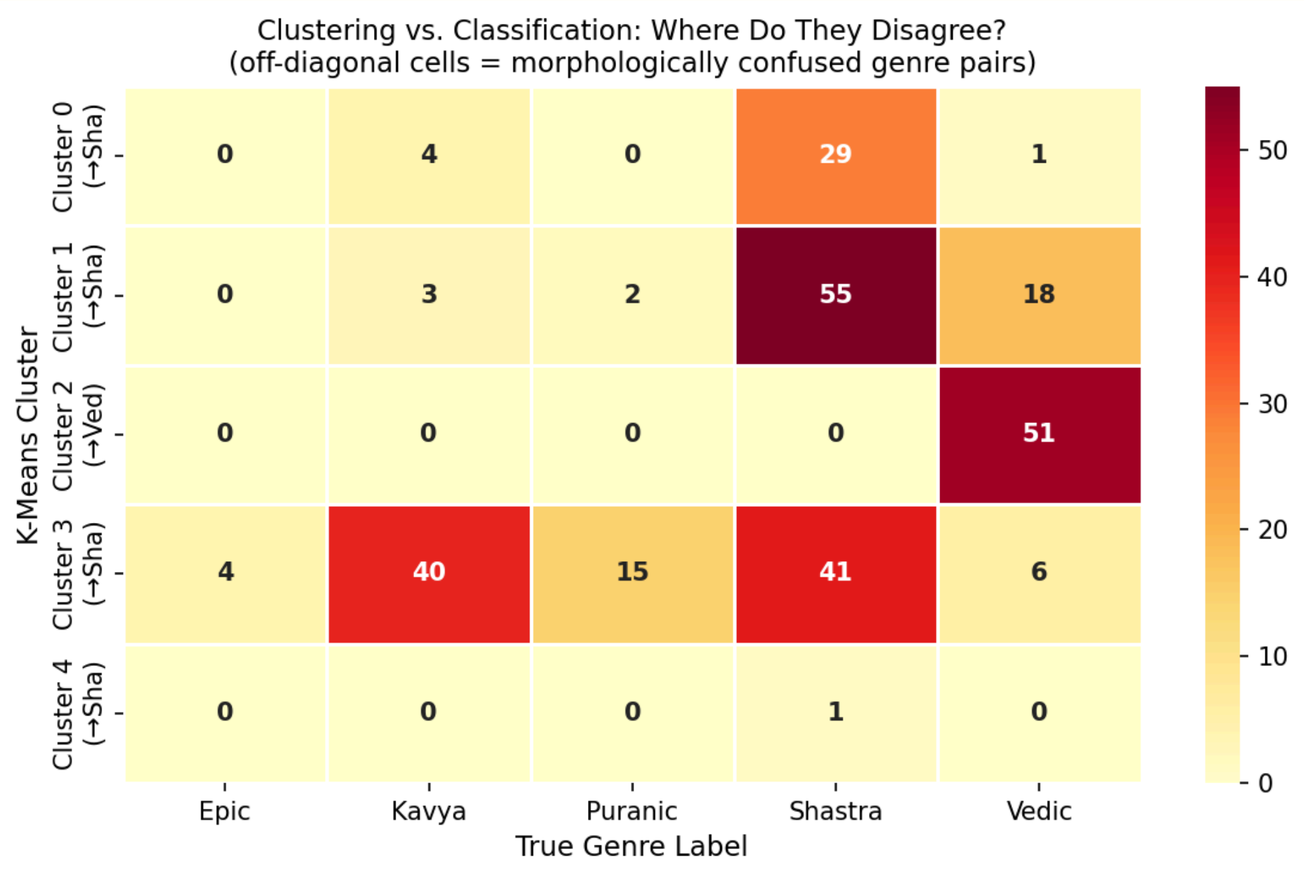
MCI by genre. Vedic has the highest median complexity (3.22 bits); Shastra the lowest (2.84 bits) with the widest spread.

- MCI ranking: Vedic > Epic > Kavya ≈ Puranic > Shastra.
- Vedic uses the full Sanskrit morphological inventory (8 cases, dual number,optative/subjunctive/injunctive moods, augmented past).

Genre	MCI	Cpd	Mantra	NOUN	NVR
Vedic	3.20	80.3	87.8	381.9	3.19
Epic	3.12	147.5	0.0	427.8	3.32
Kavya	3.03	241.4	0.0	444.0	4.01
Puranic	3.03	162.6	0.0	439.7	4.02
Shastra	2.84	236.5	0.0	502.5	7.04

NVR = nominal-to-verbal ratio; Cpd = compound rate

Clustering



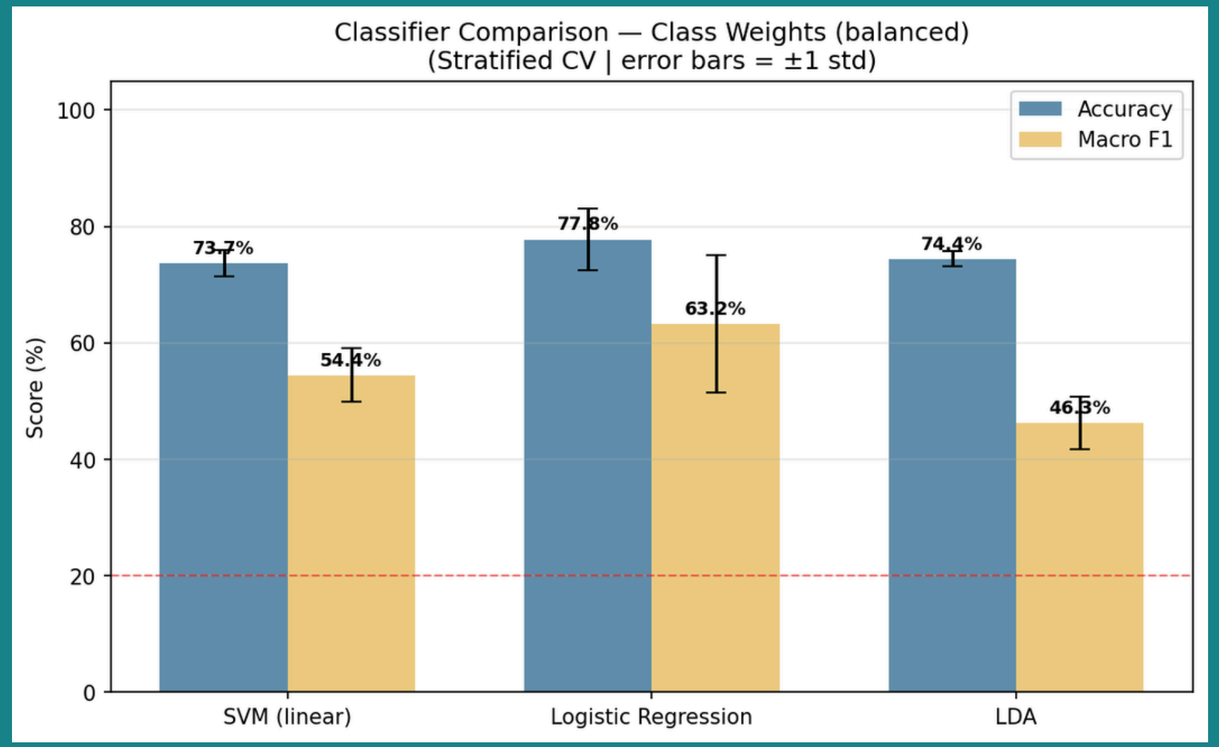
Cluster 2 recovers Vedic with 100% purity. Cluster 3 is a mixed zone containing 40/47 Kavya, 15/17 Puranic, and 41/126 Shastra texts—the primary site of morphological genre ambiguity.

- Cluster 2 recovers Vedic with perfect purity (51/51), directly **confirming Hypothesis 3**, which states Vedic texts will form a morphologically isolated cluster due to archaic case usage and verbal morphology distinct from Classical Sanskrit.
- Overall K-means achieved ARI = 0.210, indicating substantially below-chance genre recovery for non-Vedic genres.
- The remaining four clusters are all Shastra-dominant, reflecting Shastra's numerical dominance (126/270 texts) and morphological overlap with other genres.

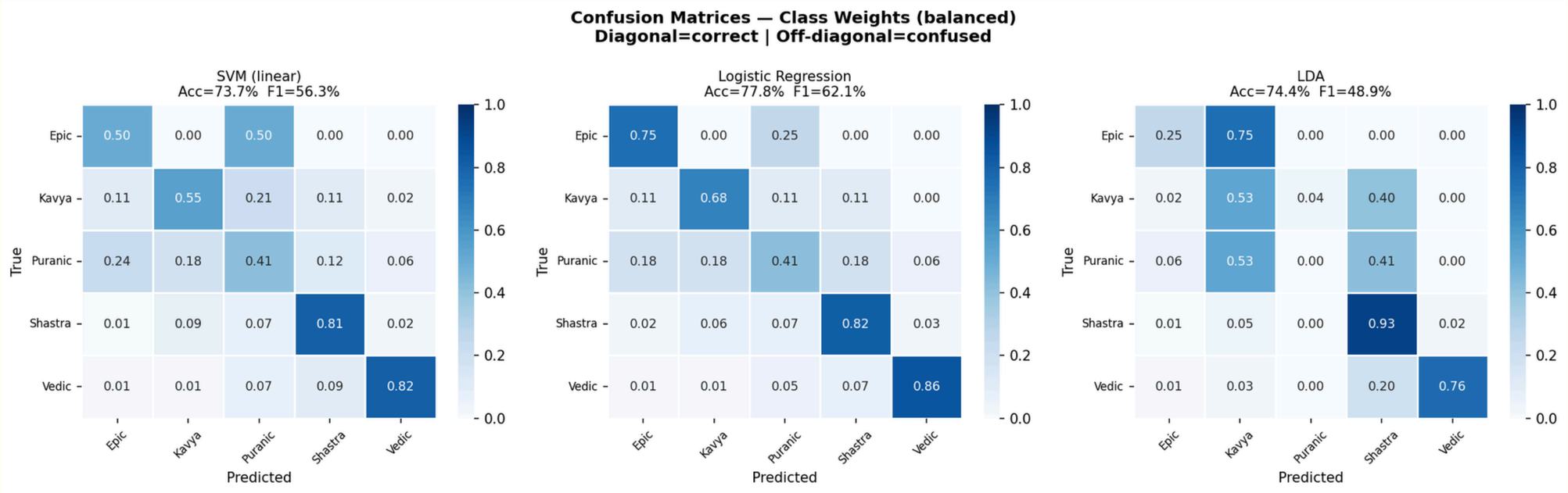
Classification

Genre	SVM	LR	LDA
Epic	0.50	**0.75**	0.25
Kavya	0.55	**0.68**	0.53
Puranic	0.41	**0.41**	0.00
Shastra	**0.81**	**0.82**	**0.93**
Vedic	0.82	**0.86**	0.76
Macro F1	54.4%	**63.2%**	46.3%

Per-genre recall with class balanced classifiers



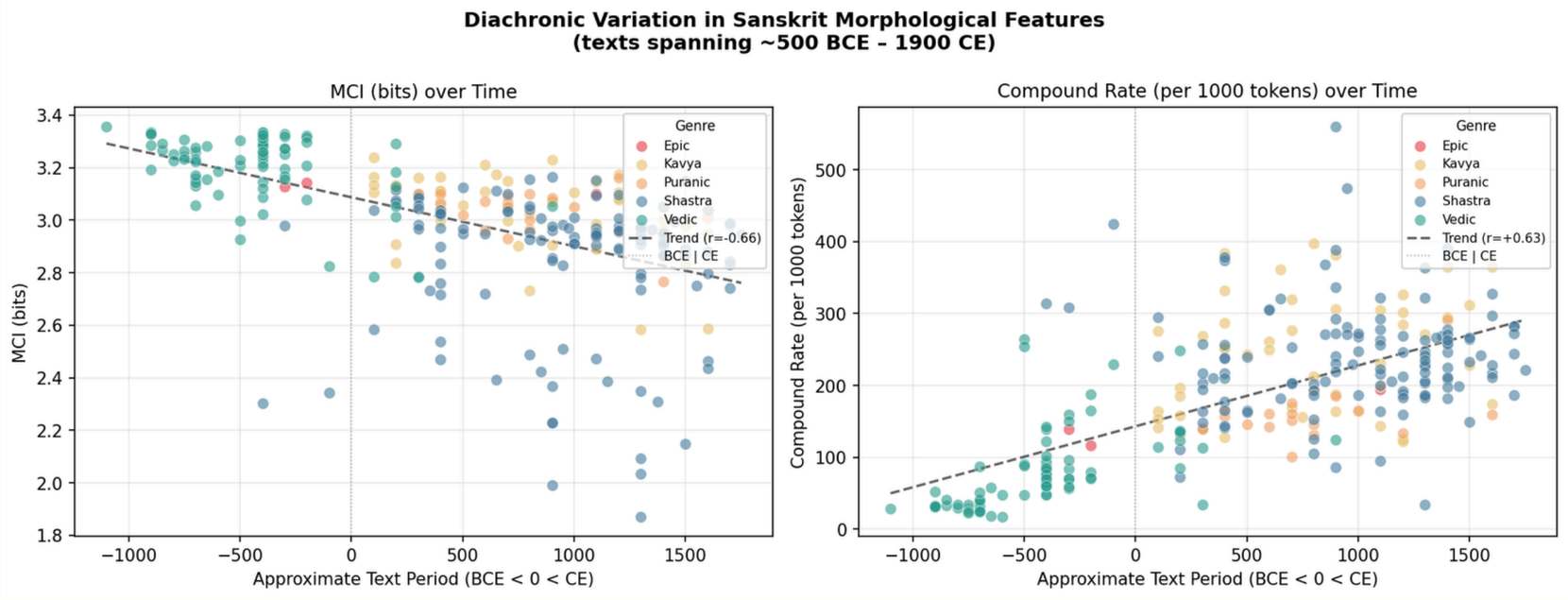
- Logistic Regression achieves the highest macro F1 (63.2%) with balanced class weights, partially disconfirming Hypothesis 4 (which predicted SVM superiority).
- The accuracy–F1 gap for SVM (73.7% accuracy vs. 54.4% F1) reveals that high accuracy is partly an artefact of the 47% Shastra baseline.



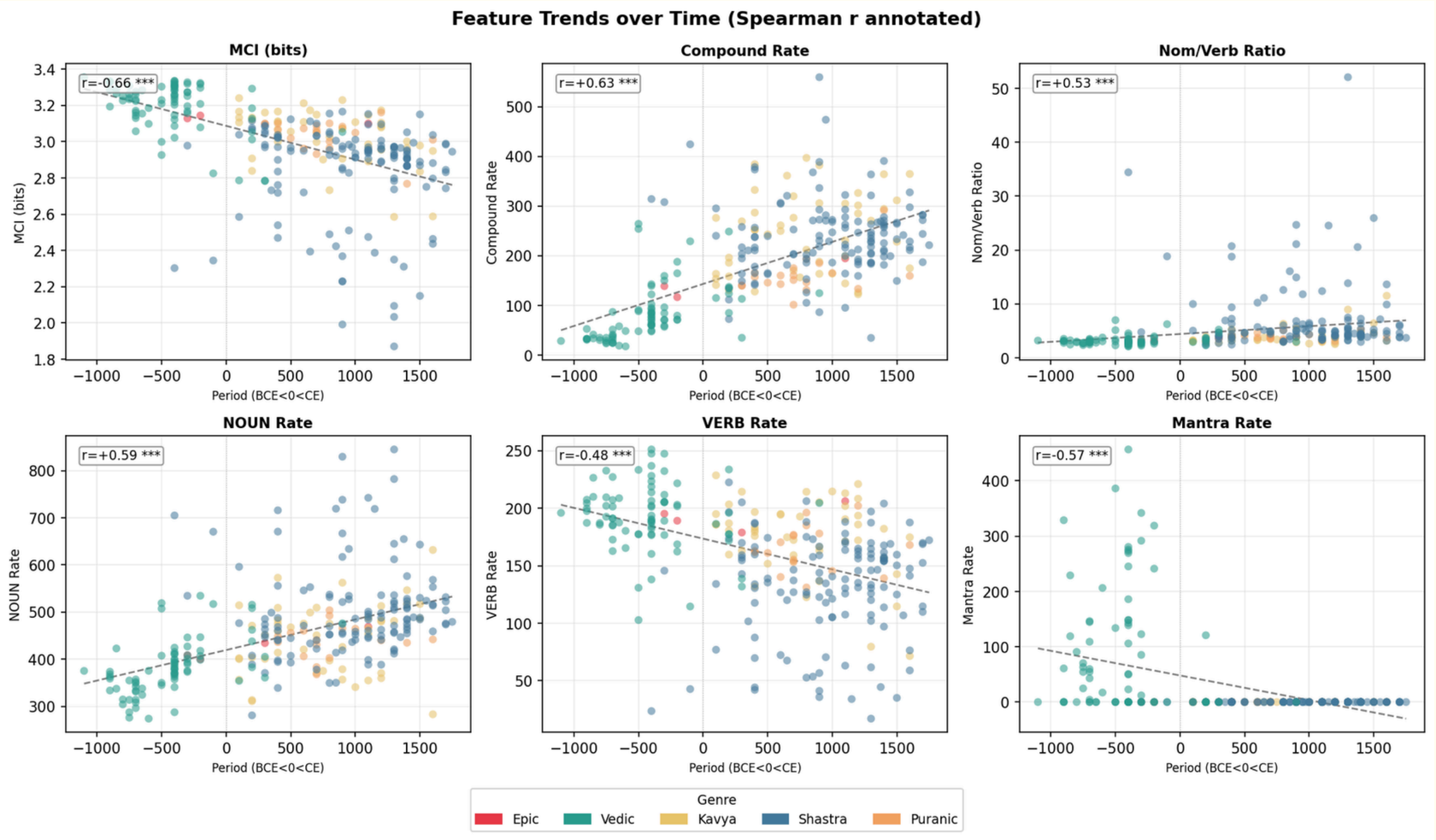
- The most important confusion pattern is the bidirectional Kavya–Shastra confusion: Kavya is predicted as Shastra by 9.7 texts on average, and Shastra is predicted as Kavya by 8.3 texts. The bidirectionality confirms that this is not a classifier bias artefact but a genuine morphological overlap.

- LDA performs worst (F1 = 46.3%) and achieves zero Puranic recall, reflecting its assumption of equal-covariance Gaussian distributions—violated by Shastra's enormous internal spread

Diachronic Spearman Correlations



Diachronic variation in MCI and compound rate. MCI declines monotonically from Vedic texts (1100 BCE) to late medieval Shastra (1750 CE; $r = -0.66$, $p < 0.0001$). Compound rate increases correspondingly ($r = +0.63$, $p < 0.0001$), suggesting a compensatory typological trade-off.

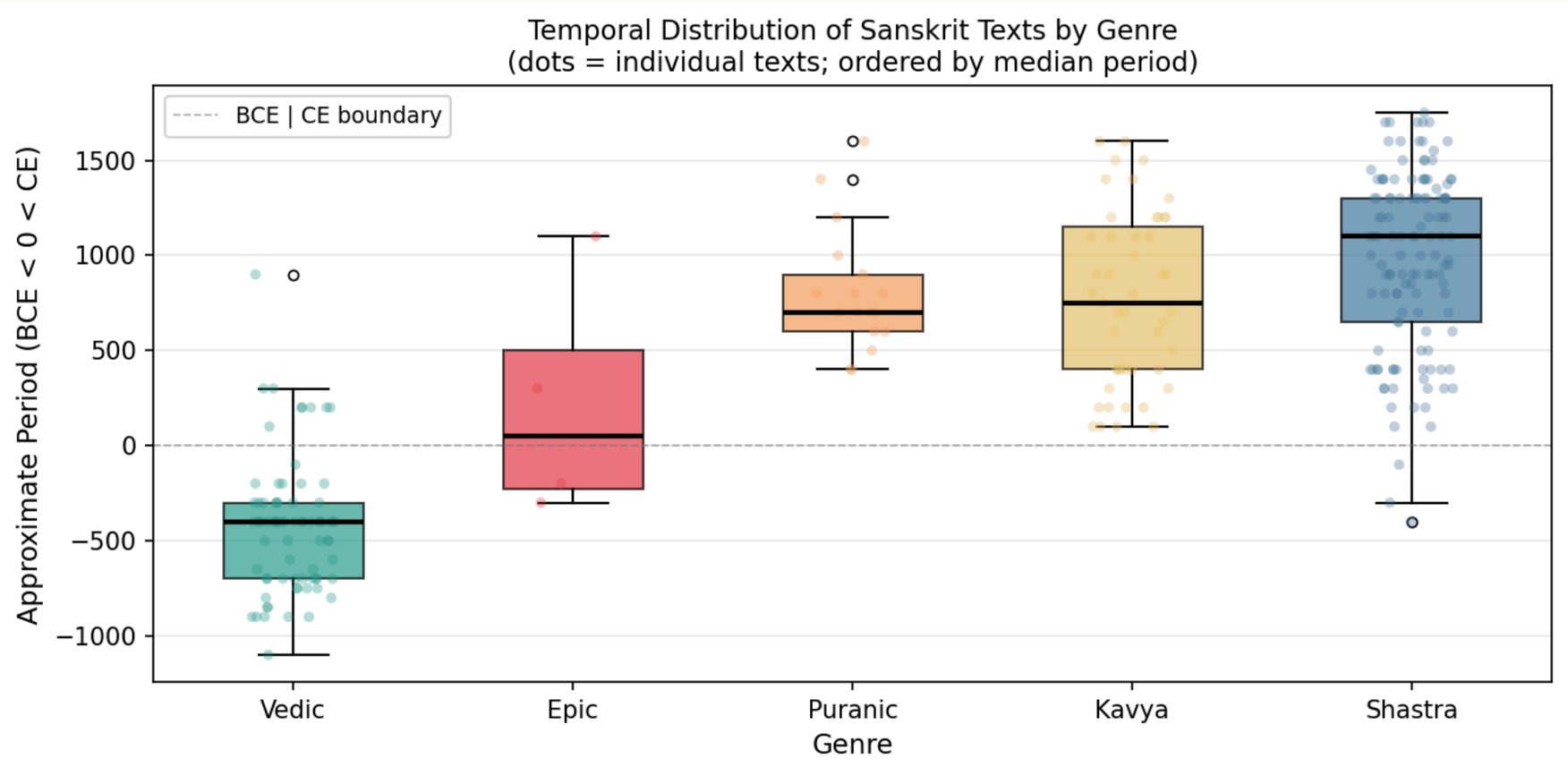


Feature trends over time. NOUN rate and compound rate increase; VERB rate, pronoun rate, MCI, and mantra rate decrease.

- Within-genre correlations show that the diachronic trend is not merely a genre-level confound: MCI declines significantly within Kavya, Shastra, and Vedic individually, indicating that the trend operates at both the genre and sub-genre level

Spearman correlations between morphological features and text period (*** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$).

Feature	τ	Sig.
MCI	−0.661	***
Compound Rate	+0.634	***
NOUN Rate	+0.589	***
Mantra Rate	−0.568	***
Nom/Verb Ratio	+0.525	***
PRON Rate	−0.507	***
VERB Rate	−0.484	***



An important methodological caveat: Shastra's lower mean MCI is partly a temporal artefact—most Shastra texts are medieval (median +1050 CE), while Vedic texts are ancient (median –450 BCE). Genre and period are confounded in this corpus, and both explanation contribute to the observed differences

Discussion

- Hypothesis 1 and 3 confirmed; Hypothesis 2 is not confirmed; Hypothesis 4 partially disconfirmed.

- **Kavya–Shastra Boundary**: Classical Sanskrit prose Kavya and technical Shastra prose share high compound density, high nominalization, and low verbal diversity. The Kavya–Shastra distinction is ultimately a content distinction (aesthetic narrative vs. technical exposition) more than a morphological one.

- Genre as temporal confound

To Conclude

1. PCA reveals a dominant nominalization axis separating the verbal Vedic register from Classical Sanskrit genres.

2. Vedic is the only genre forming a coherent morphological cluster (ARI Cluster 2: 100% pure), while Kavya, Puranic, Epic, and Shastra are morphologically interleaved.

3. Balanced-weight Logistic Regression achieves 63.2% macro F1—a substantial improvement over the 20% chance baseline but below 70%, reflecting genuine morphological ambiguity at genre boundaries.

4. Morphological complexity declines significantly across Sanskrit's 2,400-year history while compounding increases, consistent with cross-linguistic typological expectations

Thank You!!